

Zackery Fite

CS-300

Southern New Hampshire University

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Module 4 Milestone: Pseudocode

Load the Data into the Hash Table:

STRUCT Course

 courseNumber

 courseTitle

 prerequisites (list)

END STRUCT

FUNCTION loadCourses(fileName, hashTable)

 OPEN fileName

 IF file fails to open

 PRINT "Error opening file"

 RETURN

 END IF

 CREATE empty list courseNumbers

 FOR EACH line in file

SPLIT line into tokens by comma

IF number of tokens < 2

 PRINT "Invalid format"

 CONTINUE

END IF

CREATE Course object

course.courseNumber = tokens[0]

course.courseTitle = tokens[1]

ADD tokens[0] to courseNumbers

FOR i from 2 to end of tokens

 ADD tokens[i] to course.prerequisites

END FOR

INSERT course into hashTable using courseNumber as key

END FOR

FOR EACH course in hashTable

 FOR EACH prereq in course.prerequisites

 IF prereq NOT in courseNumbers

 PRINT "Invalid prerequisite: " + prereq

 END IF

 END FOR

END FOR

CLOSE file

END FUNCTION

Search and Print Course (Hash Table)

FUNCTION searchCourse(hashTable, courseNumber)

 course = hashTable.search(courseNumber)

 IF course exists

 PRINT course.courseNumber, course.courseTitle

 IF prerequisites list is empty

 PRINT "No prerequisites"

 ELSE

 FOR EACH prereq in course.prerequisites

 PRINT prereq

 END FOR

 END IF

ELSE

 PRINT "Course not found"

END IF

END FUNCTION

Print All Courses

```
FUNCTION printAllCourses(hashTable)
```

```
    FOR EACH bucket in hashTable
```

```
        FOR EACH course in bucket
```

```
            PRINT course.courseNumber, course.courseTitle
```

```
        END FOR
```

```
    END FOR
```

```
END FUNCTION
```